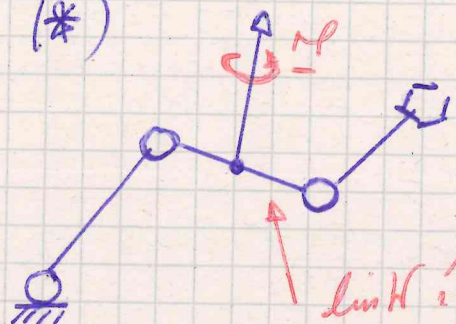


VIRTUAL WORK (*)

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MOMENTS (**)



$$\begin{aligned} \delta L &\triangleq \underline{M} \cdot \delta \theta_i = \underline{M} \cdot \sum_{j=1}^i \underline{J}_{i/j}^A \delta q_j = \\ &= \sum_{j=1}^i \left(\underline{J}_{i/j}^A \cdot \underline{M} \right) \delta q_j = \sum_{j=1}^i \left({}^0 \underline{J}_{i/j}^{AT} \cdot \underline{M} \right) \delta q_j = \\ &= \delta \underline{q}^T \begin{bmatrix} {}^0 \underline{J}_{i/1}^{AT} \underline{M} \\ {}^0 \underline{J}_{i/2}^{AT} \underline{M} \\ \vdots \\ \vdots \end{bmatrix} = \delta \underline{q}^T {}^0 \underline{J}_{i/0}^{AT} \underline{M} \end{aligned}$$

NOTE ${}^0 \underline{J}_{i/k}^{AT} \cdot \underline{M} = {}^b \underline{J}_{i/k}^{AT} \cdot {}^b \underline{M} \quad \forall {}^b$

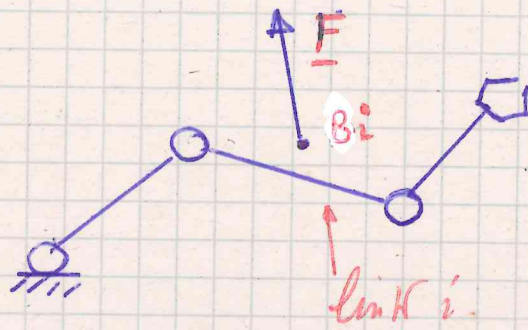
projection of $\underline{M}$
onto the space of
joint coordinates

then the projections and hence
the virtual work are invariant w.r.t. the basis
where vectors are represented

(**) Recall that moments are constant w.r.t. point
where they are expressed on a given body—

(*) The work of a force acting on a mass is the scalar
product of the force by the mass displacement. The work
of a momentum is the scalar product of the momentum by the
change of rotation—

FORCES (*)



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↑
virtual work

$$\begin{aligned}\delta L &= \underline{F} \cdot \delta \underline{r}_{B_i/o} = \underline{F} \cdot \sum_j^i \underline{J}_{B_i/j}^L \delta q_j = \\ &= \sum_j^i (\underline{J}_{B_i/j}^L \cdot \underline{F}) \delta q_j = \sum_j^i ({}^o \underline{J}_{B_i/j}^{LT} \cdot \underline{F}) \delta q_j = \\ &= \delta \underline{q}^T \begin{bmatrix} {}^o \underline{J}_{B_i/1}^{LT} \cdot \underline{F} \\ {}^o \underline{J}_{B_i/2}^{LT} \cdot \underline{F} \\ \vdots \\ 0 \end{bmatrix} = \delta \underline{q}^T \cdot {}^o \underline{J}_{B_i/o}^{LT} \cdot \underline{F}\end{aligned}$$

↑
projection of \$\underline{F}\$
onto the space of joint
coordinates

NOTE ${}^o \underline{J}_{B_i/o} = {}^o \underline{S}_{B_i/i} \cdot {}^o \underline{J}_{i/o}$

(it is invariant w.r.t.
basis where \$\underline{F}\$ and \$\underline{J}_{B_i/o}\$
are represented)

not convenient →

$$\begin{bmatrix} {}^o R & 0 \\ 0 & {}^o R \end{bmatrix} {}^i \underline{S}_{B_i/i} \begin{bmatrix} {}^i R & 0 \\ 0 & {}^i R \end{bmatrix}$$

↑
constant in \$i\$

lot of useless
computations

(*) Recall the most "important" external force
acting on a link is the "weight force" ... is always there
expressed at the center of mass \$c_i\$.

NOTE

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$$\underline{J}_{i/j}^A \cdot \underline{M} = \begin{cases} 0 & j \in TJ \\ \underline{K}_j \cdot \underline{M} & j \in RJ \end{cases}$$

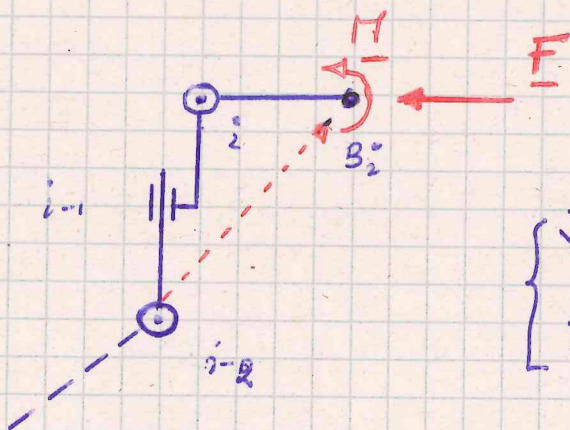
← torques do not affect TJ s

↑
orthog. projection of $\underline{M}$ along the direction of $\underline{K}_j$

$$\underline{J}_{i/j}^L \cdot \underline{F} = \begin{cases} \underline{K}_j \cdot \underline{F} & j \in TJ \\ \underbrace{(\underline{K}_j \times \underline{z}_{i/j}) \cdot \underline{F}}_{(\underline{z}_{i/j} \times \underline{F}) \cdot \underline{K}_j} & j \in RJ \end{cases}$$

$$(\underline{z}_{i/j} \times \underline{F}) \cdot \underline{K}_j$$

← projection of the moment of $\underline{F}$ w.r.t. P_j along direction $\underline{K}_j$



Compute

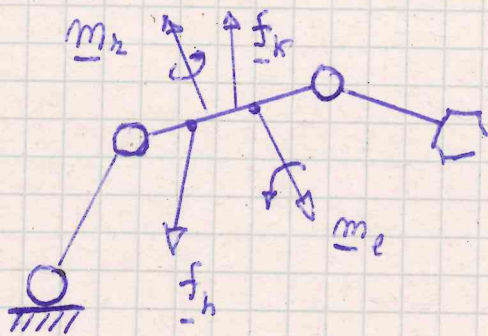
$$\begin{cases} \underline{J}_{i/i-2}^L \cdot \underline{F} \\ \underline{J}_{i/i-2}^A \cdot \underline{M} \end{cases}$$

$$\begin{cases} \underline{J}_{i/i}^L \cdot \underline{F} = 0 & \underline{z}_{i/i} \parallel \underline{F} \\ \underline{J}_{i/i}^A \cdot \underline{M} = \underline{M} & \underline{K}_i \parallel \underline{M} \end{cases}$$

$$\begin{cases} \underline{J}_{i/i-1}^L \cdot \underline{F} = 0 & \underline{K}_{i-1} \perp \underline{F} \\ \underline{J}_{i/i-1}^A \cdot \underline{M} = 0 \end{cases}$$

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TOTAL VIRTUAL WORK OF MOMENTS AND FORCES



$$\delta L_i = \sum_k \underline{f}_k \cdot \delta \underline{z}_{P_k/i} + \sum_h \underline{m}_h \cdot \delta \underline{\theta}_i =$$

$$\delta \underline{z}_{i/o} + \delta \underline{\theta}_i \times \underline{z}_{P_k/i}$$

$$= \sum_k \underline{f}_k \cdot [\delta \underline{z}_{i/o} + \delta \underline{\theta}_i \times \underline{z}_{P_k/i}] + (\sum_h \underline{m}_h) \cdot \delta \underline{\theta}_i$$

$$= (\sum_k \underline{f}_k) \cdot \delta \underline{z}_{i/o} + \sum_k (\underline{z}_{P_k/i} \times \underline{f}_k) \cdot \delta \underline{\theta}_i + (\sum_h \underline{m}_h) \delta \underline{\theta}_i$$

$$\underline{F}_i$$

$$= \underline{F}_i \cdot \delta \underline{z}_{i/o} + \underbrace{\left[\sum_k (\underline{z}_{P_k/i} \times \underline{f}_k) + \sum_h \underline{m}_h \right]}_{\underline{M}_i} \cdot \delta \underline{\theta}_i$$

$$\underline{M}_i$$

Then

$$\delta L_i = \underline{F}_i \cdot \delta \underline{z}_{i/o} + \underline{M}_i \cdot \delta \underline{\theta}_i$$

virtual work
of the resultant
of the forces applied
at P_i

virtual work of the
moments ^{applied} w.r.t.
to point P_i

Then since ${}^0(\delta \underline{z}_{i/o}) = {}^0 \underline{J}_{i/o}^L \delta \underline{q}$

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$${}^0(\delta \underline{\theta}_i) = {}^0 \underline{J}_{i/o}^A \delta \underline{q}$$

we have

$$\delta L_i = \delta \underline{q}^T \left[{}^0 \underline{J}_{i/o}^{L^T} {}^0 \underline{F}_i + {}^0 \underline{J}_{i/o}^A \underline{M}_i \right] = \delta \underline{q}^T {}^0 \underline{J}_{i/o}^T \begin{bmatrix} {}^0 \underline{M}_i \\ {}^0 \underline{F}_i \end{bmatrix}$$